

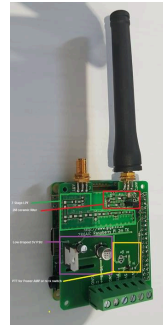
APRIL 25, 2018

CREATING A HIGH ALTITUDE BALLOON TELEMETRY SYSTEM WITH RASPBERRY PI, RPITX AND RTL-SDR

Over on his blog ZR6AIC explains how he's created a full [HAB \(high altitude balloon\) telemetry transmit and receive system using RPiTX and an RTL-SDR dongle](#) running on a Raspberry Pi 3.

RPiTX is software that enables the Raspberry Pi to transmit any modulated signal over a wide range of frequencies using just a single GPIO pin. However, the transmission contains multiple harmonics and thus requires sufficient filtering in order to transmit legally within the 2M ham band. To solve this ZR6AIC uses a [2M Raspberry Pi Hat kit which he designed and created](#) that contains two low pass filters as well as the option for an additional power amplifier.

The rest of ZR6AIC's post explains how his HAB telemetry system combines the Raspberry Pi 3, RPiTX 2M Hat, RTL-SDR, a GPS unit, battery, temperature sensor and optional camera into a full HAB transmitting system. He also explains the software and terminal commands that he uses which allows him to transmit via RPiTX a CW beacon and GPS and temperature sensor APRS telemetry data with the Direwolf software. Full instructions on setting up the also-loopback audio routing is also provided.



The 2M TX Filter by ZR6AIC



Launching the High Altitude Balloon.



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Written by [admin](#) — Posted in [Amateur Radio](#), [RTL-SDR](#) — Tagged with [APRS](#), [HAB](#), [high altitude balloon](#), [rpitx](#), [rtl-sdr](#), [rtl2832u](#), [rtl2832](#)



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AD5NL

8 years ago

FYI re: the pi hat — R278 comes out to about \$20 (US dollars). That seems like a very reasonable price... but I am not sure if Giga will ship to the U.S. (trying to get a quote for shipping got me an option for using my own shipping company?!). It would be really nice if Anton partnered with a U.S. firm to manufacture/sell/distribute such hats over here.

What's cool about this is that it actually has two band filters on it — 2 meter ceramic as well as pads for 7-element Chebyshev low pass filter. I think you will need to populate the pads for the second filter, but it should allow one to both use (say) 2 meter APRS as well as 30 meter WSPR.

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WHAT IS RTL-SDR

The RTL-SDR is an ultra cheap software defined radio based on DVB-T TV tuners with RTL2832U chips. The RTL-SDR can be used as a wide band radio scanner. It may interest ham radio enthusiasts, hardware hackers, tinkerers and anyone interested in RF.